AI-Generated Song Detection & Attribution

Technical Report

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Task	Pairwise similarity metric for AI-generated music attribution
Approach	Chunk-based feature extraction with weighted similarity scoring
Datasets	MIPPIA SMP, SONICS (97k+), FakeMusicCaps (55k clips)
Key result	F1 = 0.936 at threshold 0.90 on MIPPIA evaluation (51 pairs)

1. Problem Statement

The goal is to design a system that ingests two full-length audio tracks and outputs:

- **Attribution Score (0–1)** — the likelihood that Track B is derived from Track A (e.g., an AI-generated cover, plagiarised remix, or hallucinated generation).
- **AI Detection Score (0–1)** — the likelihood that a given track is AI-generated (produced by systems like Suno or Udio).

This goes beyond binary classification. The attribution task requires **relational modelling** robust to tempo variations, arrangement changes, and spectral artifacts introduced by modern AI generators. The detection task requires distinguishing subtle statistical differences between human recordings and machine-generated audio.

2. Solution Architecture

The system follows a four-stage pipeline: audio loading, chunked feature extraction, pairwise similarity computation, and score interpretation. Both tracks are processed identically through the first two stages before being compared in the similarity engine.

Pipeline overview

Stage	Component	Description
1. Ingestion	librosa.load()	Load audio, resample to 22,050 Hz mono
2. Chunking	Overlapping windows	30s windows with 10s hop — covers entire track
3. Extraction	Feature extractor	Per-chunk: MFCCs, Mel, Chroma, Tempo, HNR, Phase, CLAP*
4a. Attribution	Similarity engine	Best-match per chunk → weighted cosine + scalar similarity
4b. AI Detection	Logistic Regression	Trained classifier (CV F1 = 0.85) on 255 labelled tracks

**CLAP embeddings are optional — enabled via --clap flag. Requires ~1 GB model download.*

3. Feature Engineering

Feature selection rationale

Feature selection balances three concerns: (a) capturing timbral and melodic similarity for attribution, (b) detecting AI-specific spectral artifacts, and (c) keeping extraction fast enough for full-length tracks.

Feature	Dim	Rationale
MFCCs (mean + std)	80	Timbral texture — primary fingerprint of instrument/voice identity
Log-Mel Spectrogram	128	Frequency content summary across the full spectrum
Chroma (mean)	12	Pitch-class profile — captures melody and harmonic structure
Tempo (BPM)	1	Rhythmic matching; AI covers often preserve the original tempo
HNR	1	Harmonic-to-Noise Ratio: AI generators produce characteristically extreme values
Spectral Flatness	1	Near-zero in AI output (overly tonal); higher variance in human recordings
Phase Discontinuity	1	AI vocoders introduce unnatural phase jumps in the STFT domain
CLAP embedding*	512	Semantic-level audio representation; strongest single attribution signal

Are standard features sufficient?

Standard features (MFCCs, chroma) capture timbre and harmony but miss AI-specific artifacts. The inclusion of HNR, spectral flatness, and phase discontinuity addresses this gap. Empirical analysis on

MIPPIA (human) vs SONICS/FakeMusicCaps (AI) confirmed that **spectral_flatness is the strongest single discriminator** between human and AI audio (Logistic Regression coefficient: −3.64).

Attribution feature weights

Feature group	Weight	Role
CLAP embedding	0.35	Semantic similarity — dominant signal when available
MFCCs	0.15	Timbre fingerprint
Chroma	0.15	Melody and harmony
Log-Mel	0.10	Frequency content
HNR	0.08	AI artifact indicator
Spectral Flatness	0.07	AI artifact indicator
Tempo	0.05	Rhythmic alignment
Phase Discontinuity	0.05	AI artifact indicator

When CLAP is unavailable, weights are renormalised across the remaining features.

4. Handling Variable-Length Audio & Window-Bias Prevention

A naive approach analyses only the first 30 seconds — introducing **window-bias** that favours tracks with similar intros regardless of overall content. This system prevents it through two mechanisms:

- **Overlapping chunking:** Audio is segmented into 30-second windows with a 10-second hop, covering the entire track. A 3-minute track produces ~15 chunks. A final partial chunk is retained if it exceeds 10 seconds.
- **Best-match alignment strategy:** For each chunk in Track A, the system finds the best-matching chunk across all chunks of Track B (cross-track maximum). The final attribution score is the mean of all best matches.

This approach handles:

- **Segment reordering** — AI may rearrange song sections (verse/chorus order)
- **Tempo shifts** — matching chunks may be offset in time
- **Intro/outro differences** — unmatched sections do not penalise the overall score

5. AI Detection — Trained Classifier

Motivation

A purely heuristic approach (fixed Gaussian profiles) is fragile and not generalisable. To provide a statistically grounded estimate, a Logistic Regression classifier was trained on labelled audio features. Logistic Regression was chosen for its interpretability, calibrated probability outputs, and fast inference — appropriate given the modest training set size.

Training data

Source	Label	Count
MIPPIA SMP tracks	Human	154
FakeMusicCaps (5 TTM models)	AI	100
SONICS (Suno)	AI	1
Total		**255**

Features used

`spectral_flatness, zcr_mean, mfcc1_mean, spectral_centroid_hz, hnr_db, tempo_bpm, phase_discontinuity`

Results (5-fold stratified cross-validation)

Metric	Score
Accuracy	0.867 ± 0.049
Precision	0.767 ± 0.060
Recall	**0.960 ± 0.080**
F1	**0.851 ± 0.057**

High recall (0.96) is the most important property for this use case — the system is unlikely to miss an AI-generated track.

Feature importance

`spectral_flatness` dominates with a coefficient of −3.64, confirming that AI generators produce abnormally low spectral flatness compared to human recordings. `zcr_mean` and `mfcc1_mean` contribute secondary discriminative power.

Important caveat

The AI detection score is a probabilistic estimate, not a proof. Scores are reported as likelihood values (e.g., “likely AI-generated”) rather than categorical verdicts. The model was trained on 255 samples — a larger, more diverse training set would improve both precision and generalisation.

6. Attribution — Evaluation on MIPPIA SMP

Dataset

51 complete pairs (both tracks downloadable) from the MIPPIA SMP dataset were evaluated. Pair types:

Relation	Count
plag	21
plag_doubt	16
remake	13

Score distribution (related pairs)

Statistic	Value
Mean	0.9605
Std	0.0209
Min	0.9007
Max	0.9904

Precision / Recall / F1 at multiple thresholds

Unrelated pairs were generated by cross-pairing tracks from different MIPPIA pairs (50 synthetic negative examples).

Threshold	Precision	Recall	F1	Accuracy
0.70	0.593	1.000	0.745	0.654
0.80	0.680	1.000	0.810	0.762
0.90	**0.879**	**1.000**	**0.936**	**0.931**

Recall = 1.0 at all thresholds: every related pair was correctly identified. The system trades some precision for high sensitivity — appropriate for a music attribution use case where false negatives (missing a real attribution) are more costly than false positives.

7. Datasets

Dataset	Role
MIPPIA SMP (158 pairs)	Ground-truth labelled pairs for attribution validation and human feature profiling
SONICS (97k+ tracks)	AI-generated audio (Suno/Udio) for detector training and artifact analysis
FakeMusicCaps (55k clips)	Fine-grained AI artifact analysis from 5 TTM models; classifier training data

All datasets are excluded from the repository. Download instructions are provided in README.md.

8. Tools & Libraries

Category	Library	Purpose
Audio I/O & DSP	librosa	Loading, resampling, STFT, MFCCs, chroma, beat tracking
Audio I/O	soundfile	WAV/FLAC reading
Neural embeddings	transformers	CLAP model (laion/clap-htsat-unfused)
Deep learning	torch	CLAP inference backend
ML classifier	scikit-learn	Logistic Regression, StandardScaler, cross-validation
Numerics	numpy, scipy	Array operations, signal processing
Data	pandas, datasets	Dataset loading and manipulation

Category	Library	Purpose
Visualisation	matplotlib, seaborn	Feature distribution plots
Dataset download	yt-dlp	MIPPIA audio retrieval from YouTube

9. Design Decisions & Trade-offs

Decision	Rationale	Trade-off
30s chunks, 10s hop	Covers full track; captures local patterns while maintaining global coverage	Higher compute cost vs. a single global embedding approach
Best-match (not DTW)	Simple, fast, handles segment reordering naturally	May overestimate similarity for partially similar tracks
Cosine over Euclidean	Scale-invariant; better suited for high-dimensional embeddings	Loses magnitude information
Logistic Regression	Interpretable, calibrated probabilities, fast inference	Less powerful than deep neural classifiers
CLAP as optional	System works on CPU without large downloads	Weaker genre-level attribution without neural embeddings
max_diff per scalar	Domain-adapted normalisation for each scalar metric	Requires domain knowledge to tune correctly

10. Limitations

- **AI detector training set:** 255 samples is a modest set. The classifier may not generalise well to AI generators not represented in FakeMusicCaps/SONICS. Retraining on the full SONICS dataset (97k tracks) would substantially improve robustness.
- **Attribution ≠ AI detection:** A high attribution score indicates the tracks are musically related — it does not imply that either track is AI-generated. These are two separate tasks with separate outputs.
- **CLAP dependency:** The strongest attribution feature requires ~1 GB model download and benefits from GPU acceleration. Without CLAP, the system relies entirely on spectral features.
- **No tempo normalisation:** Chunks are compared at their original BPM. A significantly sped-up or slowed-down cover may produce a lower attribution score.
- **Stereo → mono collapse:** Stereo panning information is discarded during preprocessing.
- **YouTube availability:** MIPPIA audio is sourced from YouTube — availability varies over time, limiting reproducibility.
- **Best-match inflation:** The max-pooling strategy can inflate scores when one strong chunk in B matches many chunks in A. A symmetric matching approach (e.g., Hungarian algorithm) would mitigate this.

11. Future Improvements

- **Larger training set:** Retrain AI detector on the full SONICS dataset (97k tracks) or augment with additional generators (MusicGen, Stable Audio).
- **Dynamic Time Warping (DTW):** Replace best-match chunk alignment with DTW for more principled handling of time-stretched covers.
- **MERT embeddings:** Replace or complement CLAP with the music-specific MERT model for richer semantic representations.
- **Tempo normalisation:** Automatically detect BPM ratio between tracks and align chunks before comparison.
- **Explainability:** Add SHAP values or per-feature contribution heatmaps to explain why a particular score was assigned.
- **End-to-end fine-tuning:** Fine-tune CLAP on MIPPIA pairs as a contrastive learning task to learn attribution-specific representations.
- **Symmetric matching:** Replace one-directional best-match with mutual best-match or Hungarian algorithm to reduce score inflation.

12. Usage

Basic comparison (no CLAP):

```
python compare_tracks.py original.mp3 suspected_cover.mp3
```

With CLAP neural embeddings (more accurate):

```
python compare_tracks.py original.mp3 suspected_cover.mp3 --clap
```

JSON output:

```
python compare_tracks.py original.mp3 suspected_cover.mp3 --json
```

Train the AI detector (run once after downloading datasets):

```
python train_ai_detector.py
```

Evaluate on full MIPPIA dataset:

```
python evaluate_mippia.py
```

Compute precision/recall/F1:

```
python evaluate_attribution.py
```

See `data_exploration.ipynb` for an end-to-end walkthrough using real audio from the MIPPIA dataset and SONICS.